# Urban Scale Digital Twins Vis-à-Vis Complex Phenomena

## Datafication and Social and Environmental Equity

*Marianna Charitonidou*
*Athens School of Fine Arts*
*m.charitonidou@icloud.com*

*The paper analyses the role of the urban scale digital twins in conceiving and designing urban spaces. Urban scale digital twins are virtual replicas of cities. Within the current context of data-driven societies, they are often used to test scenarios related to sustainable environmental design. An important reorientation concerning the role of the urban scale digital twins in urban planning is that from technical to the socio-technical perspectives. The debates on Smart Cities often focus on technical issues, neglecting the social issues concerning urban planning. At the core of the paper is the idea that in order to combine environmental equity and social equity, it is pivotal to bring together the social and the technical viewpoints of urban planning. This can become possible through the adoption of socio-technical perspectives. The paper explores the role of complexity perspectives based on new types of urban data in reshaping urban planning decision making methods. It pays special attention to datafication, Smart Cities, digital exclusion, Big Data, IoT, AI, Machine Learning, automated data collection, and data analytics. It also investigates how urban scale digital twins can measure, describe and analyze complex phenomena. Particular emphasis is placed on how urban scale digital twins help develop new data driven scenarios, promote sustainable development goals, and shape new participatory design methods. At the center of the paper are the following main characteristics of urban scale digital twins: firstly, their 'scalability'; secondly, their 'predictability', which becomes possible thanks to the use of simulation algorithms; thirdly, their capacity to integrate new elements thanks to the use of IoT sensors, and data undated concerning in situ real-time data, and, finally, their capacity to enhance cooperation due to the fact that they can be broadly accessible.*

**Keywords**: *Social Equity, Environmental Equity, Big Data, IoT, AI, Machine Learning, Automated Data Collection, Data Analytics*

## INTRODUCTION

The term 'digital twin' refers to the digital representation enabling comprehensive data exchange and can contain models, simulations and algorithms describing their counterpart and its features and behavior in the real world[1]. A 'digital twin' is a digital representation of a physical process, person, place, system or device. The term 'digital twin' firstly emerged in the field of manufacturing sector to refer to digital simulation models that run alongside real-time processes. 'Digital twins' are digital replicas of physical entities. Their creation is based on the use of advanced technological applications, such as sensing, processing, and data transmission. Digital twins are used in the field of urban analytics, as well as in the field of computational social sciences.

Recently, within the field of smart cities, the notion of urban scale digital twin has acquired a central place (Charitonidou 2022).

Digital twins enhance evidence-based operational decisions and experimentation on urban policies. The current state of research concerning the role of digital twins in shaping urban policies is characterized by a dichotomy between scholars that focus on the technological and sustainable benefits of the use of urban scale digital twins and researchers that criticize 'digital universalism'. This paper intends to challenge this dichotomy, shaping methods based on a socio-technical perspective of using urban scale digital twins, and combining the technical, sustainable and social advantages of their use. Li Deren, Yu Wenbo, Shao Zhenfeng argue that at the core of the development of urban-scale digital twins is the creation of "a complex giant system between the physical world and the virtual space that can map each other and interact with each other in both directions" (Deren, Wenbo, Zhenfeng, 2021, p. 1). The continuous generation of massive urban Big Data and the use of sensors within the cities for which the digital twins are created are necessary for the construction of urban scale digital twins (Deren, Wenbo, Zhenfeng, 2021, p. 1).

## URBAN SCALE DIGITAL TWINS AND SUSTAINABLE ENVIRONMENTAL DESIGN

To realize the central role of Europe within the framework of the endeavors to incorporate urban scale digital twins in decision making concerning urban planning, we should take into account the fact that "Europe is emerging as the main centre of development of urban digital twins, with over 60 % of the existing" (Ferré-Bigorra, Casals, Gangolells, 2022) urban scale digital twins. The European Union has set the following goals regarding sustainable urban planning strategies: firstly, the empowerment of "urban actors towards common goals; secondly, the development of people-oriented urban planning strategies that aim to contribute to the social equity of communities; thirdly, the development of digital platforms and other digital tools that intend to enhance interactive and proactive approaches in urban planning decision-making, and "the creation of integrated, open, and functional technological infrastructures for the development of programmes and the provision of services (data-driven planning)" (Caprari, Castelli, Montuori, Camardelli, Malvezzi, 2022). Among existing urban scale digital twins that are either in operation or under development are the twins of the following cities or districts: that of Athens in Greece, that of Plzeň in the Czech Republic, that of Dublin Docklands in Ireland, that of Herrenberg in Germany, that of Vienna in Austria, that of Zurich in Switzerland, that of New York in the United States of America, that of London in the United Kingdom, and that of Helsinki in Finland. Other note-worthy urban scale digital twins are those of Cambridge, Gothenburg, Munich, Newcastle, Paris, Rennes and Rotterdam (Caprari, Castelli, Montuori, Camardelli, Malvezzi, 2022).

Two programs that play a major role in shaping sustainable urban planning methods are the European New Green Deal, the Agenda for Sustainable Development and its Sustainable Development Goals, which is also known as SDGs. The former – the European Green Deal – is based on the intention to achieve zero net emissions by 2050. This program places particular emphasis "on achieving a circular economy by 2050, creating a sustainable food system and protecting biodiversity and pollinators. As John Hatcher remarks, "60% of organizations across major sectors are leaning on digital twins as a catalyst […] to fulfil their sustainability agenda" (Hatcher, 2022). According to Hatcher, "digital twin implementations are set to increase by 36% on average over the next five years" (Hatcher, 2022). Note-worthy the following main characteristics of urban scale digital twins: their scalability; their predictability thanks to the use of simulation algorithms; their capacity to integrate new elements thanks to the use of IoT sensors, and real-time data, and their capacity to enhance cooperation due to the fact that they can be broadly accessible (Caprari, Castelli, Montuori, Camardelli, Malvezzi, 2022). Citizens can download and upload data enhancing in

this way social equity and participatory design methods (Caprari, Castelli, Montuori, Camardelli, Malvezzi, 2022). Gordon S. Blair distinguishes three challenges concerning the creation of digital twins: firstly, the challenge of "bringing the environmental assets together in one logical place, including both data assets and modelling assets"; secondly, the challenge of allowing different assets to work together as part of a larger digital twin architecture", and, thirdly, the challenge of ensuring "that the necessary storage and processing capacity is available when it is needed, especially given the sizes of the challenges and the associated potentially very large datasets." (Blair, 2021)

## ON THE BENEFITS OF CYBER–PHYSICAL–SOCIAL SYSTEM WITH COUPLED PROPERTIES

An important shift within the field of smart cities is that from technical to socio-technical perspectives. It is related to the endeavor to reveal "multiple dimensions beyond an infrastructure-technology focus" (Joss, Sengers, Schraven, Caprotti, Dayot, 2019, p. 24). The shift from technical to socio-technical perspectives goes hand in hand with the effort to construct urban scale digital twins that aim to "reflect the specifics of the urban and socio-political context." (Nochta, Wan, Schooling, Parlikad, 2021) According to Martin Tomko and Stephan Winter, "[t]he term 'digital twin' has been applied to representations of buildings and aggregations thereof such as precincts or entire cities – as long as these representations preserved aspects of temporal dynamics and self-updating ("4D")." Tomko and Winter have criticized the term 'digital twin' (Tomko, Winter, 2019, p. 396). Their critique departs from Michael Batty's remark that "a computer model of a physical system can never be the basis of a digital twin [i.e., 'mirror'] for many elements of the real system are ignored in any such abstraction" (Batty, 2018, p. 817). Tomko and Winter, in contrast with Batty, argue that the notion of 'digital twin' should be replaced. They suggest that the notion that should replace the term 'digital twin' should be "cyber–physical–social system with coupled properties" (Tomko, Winter, 2019, p. 396). They argue that this shift is related to a recognition of the fact that digital models do not function exclusively as a "passive reflection of a mirror" (Tomko, Winter, 2019, p. 396), but most importantly as systems serving to establish methods of action.

According to Batty, "[t]he idea of the digital twin [...] has emerged from the representation of the city in terms of its physical assets." (Batty, 2018, p. 818) Batty also argues that "one of the quests in city modelling is to merge social and economic processes with the built environment and to link functional and physical processes to socio-economic representations." (Batty, 2018, p. 819) The digital twins are able to get updated following the changes of the physical equivalents thanks to the pairing between the virtual and the physical world. To understand what is the main idea behind the creation of digital twins we should bear in mind that "[a]n ideal digital twin would be identical to its physical counter-part and have a complete, real-time dataset of all information on the object/system" (White, Zink, Codecá, Clarke, 2021, p. 103064).

Among the challenges of data-driven approaches are the measurement errors, the biases, the existence false positives and false negatives, the undesired discrimination effects, the complexity, the network effects, the non-linear dynamics, the wicked problems, and an ensemble of convergence issues. Apart from the aforementioned issues, a problem that should be highlighted is the fact that, in general, the digital twin approaches have been largely ignorant of people and what relates to them. This means that the ways in which urban scale digital twins function often neglect the importance of social interactions, competition and cooperation, social norms, laws and regulations, culture, history, politics, democracy, human rights, ethics, and essential non-material qualities. It is, therefore, indispensable to develop approaches that aim to incorporate questions related to the aforementioned aspects in the ways in which urban scale digital twins are created and used. The fact that the role of urban scale digital twins in the decision-making processes concerning urban

planning will become even more important during the next years makes the incorporation of aspects related to democracy, human dignity, and solidarity in how the urban scale digital twins function even more necessary. Clare Wildfire distinguishes two categories of benefits of the city-scale digital twins: the reactive benefits, on the one hand, and the predictive benefits, on the other. Wildfire relates the first category of benefits to the capacity of enhancing "real-time or near real-time interventions and improve the smooth day-to-day running of the city or asset", and the second category to the use of data for the improvement of "longer-term scenario planning to steer appropriate (and equitable) investment decisions" (Wildfire, 2018).

## THE CASE OF THE NATIONAL DIGITAL TWIN PROGRAMME (NDTP)

The "National Digital Twin programme" (NDTp), which belongs to the programs of the Centre for Digital Built Britain is based on a collaboration between the University of Cambridge and the Department for Business, Energy and Industrial Strategy. The so-called "Gemini Principles" aim "to build consensus on foundational definitions and guiding […] and to begin enabling alignment on the approach to information management across the built environment" (Bolton., Butler, Dabson, Enzer, Evans, Fenemore, Harradence, et al., 2018). The aforementioned program was launched in July 2018. At the core of this program is the intention to contribute to society, to economy, to business and to environment. The NDTp runs the Digital Twin Hub, a collaborative and supportive web-enabled community for those who own, or who are developing digital twins within the built environment. According to the Institution of Engineering and Technology (IET), the environmental benefits of the use of urban scale digital twins should focus on the endeavors to minimize waste, to achieve resource efficiency, and to enhance circular economy in the built environment. The IET places particular emphasis on the advantages of the use of urban scale digital twins for the endeavors to minimize waste

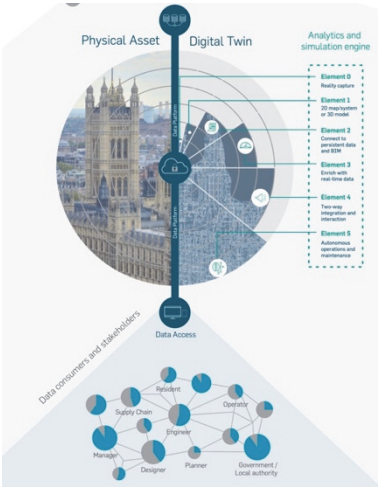


Figure 1 Digital twin maturity spectrum. Source: https://www.theiet.org/media/8762/digital-twins-for-the-built-environment.pdf (Accessed on 1 April 2022)

for economy. I could highlight its intention to "increase national productivity from higher-performing and resilient infrastructure operating as a system"[2] (figure 1). The IET also intends to conceive "[a] new smart transport or supply system might be implemented with step-change improvements to quality of life and sustainability"[3].

## THE URBAN SCALE DIGITAL TWIN OF LONDON BY ACCUCITIES

A worth-mentioning urban scale digital twin is that of London, which was created by AccuCities[4] (figure 2, figure 3). AccuCities is a London-based 3D mapping agency, established in 2017. Apart from the urban scale digital twin of London, AccuCities created models of Birmingham, Bristol, Cardiff and Dublin. Moreover, AccuCities created the Urban planning application "Plan.City". This application combines models and third-party data. Its tools include a Builder Tool, and an Area of Potential visibility. The urban scale digital twin of London is accessible through the application named "Plan.City". The datasets of the aforementioned urban scale digital twin of London are publicly available. The users of "Plan.City" are able to build a maximum building envelope using a 3D Builder Tool. Unreal Engine 4 (UE4) can be used to

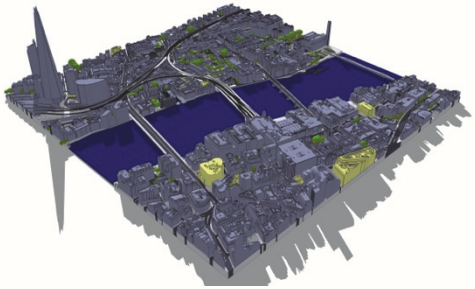

Figure 2
Image from 3D London | AccuCities. Source: https://www.accucities.com/product/tq3280-free-3d-london-samples/ (accessed 7 January 2023)

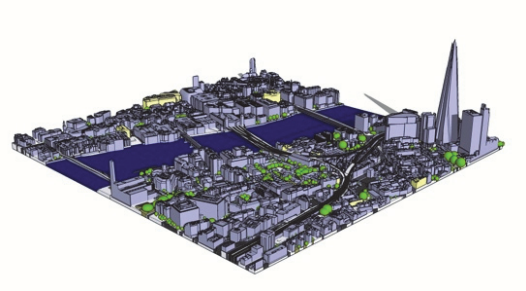

Figure 3
Image from 3D London | AccuCities. Source: https://www.accucities.com/product/tq3280-free-3d-london-samples/ (accessed 7 January 2023)

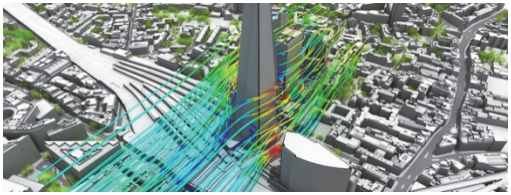

Figure 4
Wind Microclimate Analysis of London by SimScale. Source: https://www.great.gov.uk/international/trade/case-study/3474/3d-model-of-london-in-simscale-wind-comfort-simula/(accessed 7 March 2023)

assist decision-making concerning the urban environment of London. Among the analyses for which the aforementioned urban scale digital twin of London was used is that concerning wind microclimate. For this analysis, SimScale was used. SimScale takes into account data concerning use exact wind statistics, run wind analyses for 36 wind directions, and assess pedestrian wind comfort and safety according to the London LDDC criteria on an annual, seasonal, and also worst-case scenario. Additionally, SimScale can also be used to analyze the transient wind flow patterns around the buildings using streamlines, velocity vectors and much more for each of the 36 wind directions[4] (figure 4).

## THE COLLABORATION BETWEEN BOLOGNA AND BARCELONA AND URBAN ANALYTICS

In July 2022, the Barcelona Supercomputing Center-Centro Nacional de Supercomputación (BSC-CNS), the CINECA Consortium of Universities, the University of Bologna, the city council of Barcelona and the city council of Bologna signed an agreement concerning the creation of a project focusing on urban governance. This project is based on the development of urban scale digital twins applications. At the core of this collaborative project between Bologna and Barcelona is the intention to shape evidence-based decision-making strategies concerning not only public policies, but also impact assessment. This collaborative project aims to place particular emphasis on developing urban governance methods concerning parameters related to urban mobility and sustainable environmental design. More specifically, special attention will be paid to urban planning policies intending to eliminate greenhouse-gas emissions. Moreover, this project aims at the development of urban scale digital twins to be used by the municipality of Bologna as well as the municipality of Barcelona for implementing ecological public policy models in both cities. Aitor Hernández-Morales, in his article entitled "Barcelona bets on 'digital twin' as future of city planning", refers to the MareNostrum supercomputer, which is housed in Torre Girona chapel in Barcelona and is among the most powerful data processors internationally (Hernández-Morales, 2022). This supercomputer is used for the aforementioned collaborative project between Barcelona and Bologna. He also mentions that this supercomputer is expected to process data with the objective of improving urban policies.

## THE CASE OF 'DIGITAL URBAN EUROPEAN TWINS' (DUET)

Another interesting case that is based on the use of urban scale digital twins is the project entitled 'Digital Urban European Twins' (DUET), which is "a cooperative endeavour, involving 15 different partners from across Europe". Among the cities that

are involved in this project are Madrid, The Hague, Budapest, Lyon, Oslo, Porto, Athens, Antwerp and Pilsen. DUET is based on the intention to experiment with new decision-making processes, using on-demand logistics operations. Through the use of urban scale digital twins, it explores the potentials of new strategies as far as low-emission logistics operations are concerned. Urban scale digital twins render possible the integration of data and the modelling of possible strategies, a wide variety of solutions for shared, connected, and low-emission logistics operations. Among the parameters that the urban scale digital twins of the DUET project simulate are those concerning different traffic, air and noise. Particular attention is paid to the exploration of traffic models, including static, dynamic and a local mobility models (Cityflows). In order to simulate air quality data concerning traffic volume, road network, wind speed and wind direction are used. Despite the interest of the project in taking into account how citizens perceive urban landscape, and in trying to render the urban scale digital twins "citizen-centric", the efforts to "model citizens by looking at their emotional state" entail risks given that they are based on significant abstraction and simplifications. While urban scale digital twins seem very efficient in simulating parameters related to sustainable environmental design, they are rather problematic when they intend to simulate "how people experience their built environment [in order] [...] to develop smart cities in line with evolving patterns and preferences" (Raes, Michiels, Adolphi, Tampere, Dalianis, McAleer, Kogut, 2022).

## URBAN COMMONS AND URBAN SCALE DIGITAL TWINS

At the core of this paper is the exploration of how urban scale digital twins can help us use Big Data to enhance social advocacy. Incorporating urban scale digital twins in the decision-making processes concerning urban planning, urban planners can shape new participatory design methods. This could become possible through the interaction of citizens with the visualization of data concerning the existing situation of urban environments. The capacity or urban scale digital twins to visualize data and make them accessible to non-specialized audience provide an opportunity to take seriously into account the opinions of citizens concerning a broad specter of parameters during the decision-making processes. More specifically, the possibility to create applications that would visualize data concerning social and environmental equity in cities could contribute to the endeavors to combine the potentials that advanced technologies offer with the intention to promote methods of decision-making in urban planning that focus on participation. Within the framework of such efforts to bridge the opportunities that advanced technologies provide with the incorporation of the feedback of citizens during the different phases of decision-making in city planning, one should not forget the risks of 'digital universalism' (Loukissas, 2019).

Manuel Castells's approach is useful for deciphering the tension between the real and the ideal at stake during the process of abstracting sets of variables and processes in the case of urban scale digital twins. Castells argues that the societal system corresponding to the digital era is based on informationalism and globalism. He also claims that societal processes cannot be understood or represented without the underlying technology. At the core of Castells's *Communication Power*, is the following question: "where does power lie in the global network society?" (Castells, 2009, p. 42). Castells distinguishes four categories of power in the network society: networking power, network power, networked power, network-making power. He argues that the network-making power is the "paramount form of power in the network society" (Castells, 2009, p. 47), and that the network society is organized around the following three concepts: 'space of flows', 'space of places', and 'timeless time' (White, 2016, p. 1674). Castells, through these concepts, intends to render explicit that the "incorporation of the impact of advanced forms of networked communication" (White, 2016, p. 1673, 1674) calls for a new understanding of societies. According to Castells, in

network society there are no boundaries, and, for this reason, contemporary urbanization and networking dynamics should be studied conjointly. Castells's theory could help us better understand the relationship between Big Data and urban planning in data-driven society and the new kind of temporality in the so-called 'network-society' (Castells, 1989; 2009; 2010a; 2-10b; 2021). Michael Batty's and Castells's approaches are related to the transition from spatial perspectives concerning the investigation of urban data to topological perspectives (Batty, 2018; Castells, 1989; 2009; 2010a; 2010b; 2021).

Useful for addressing issues related to social and spatial mobility simultaneously are the concepts of 'motility', and 'mobility justice'. The concept of 'motility' is used by Vincent Kaufmann and is related to the intention to understand social and spatial mobility as capital, and the endeavor to address the displacement of both concrete entities, such as consumables, machinery or people, and abstract entities, such as information, ideas or norms, simultaneously (Kaufmann, 2003). Mimmi Sheller coined recently the term 'mobility justice' to respond to the dilemma whether the term migration or mobility is more socially equitable and to suggest a new way of understanding inequality and uneven accessibility to the mobility commons (Sheller, 2018). 'Mobility justice' could help us address the issue of unequal access to urban scale digital twins's data. This concept is useful for analyzing how urban scale digital twins can be used for simulations concerning urban mobility. The main idea behind the use of the term 'mobility justice' is the intention to render explicit that while mobility is a fundamental right for everyone, it is experienced unequally along the lines of gender, class, ethnicity, race, religion and age. As Sheller argues, "a strong theorization of mobility justice is the best way to bridge these various dimensions of urban inequalities" (Sheller, 2020). Sheller also underscores the fact that shaping urban planning methods that aim to promote a sustainable future for the cities should go hand in hand with using less destructive modes of urban mobility. To shape approaches that promote the use of Big Data for urban analytics without neglecting the social aspects involved in the strategies of formation of urban policies, it is important to bear in mind the weaknesses of 'digital universalism' and the assumptions on which the creation of urban scale digital twins. To do so, pivotal is the epistemological shift from technical to socio-technical perspectives.

A distinction that is at the core of the debates around participation-oriented strategies is that between the collaborative and the co-production approaches. The distinction between the collaborative and the co-production approaches is examined by Vanessa Watson remarks that, "co-production" and "collaborative or communicative planning," despite their shared concern "with how state and society can engage in order to improve the quality of life of populations […] with an emphasis on the poor and marginalized," differ in the sense that co-production "works outside (and sometimes against) established rules and procedures of governance in terms of engagement with the state, while this is much less usual (although not impossible) in collaborative and communicative planning processes" (Watson, 2014, p. 71). As Watson argues, 'negotiated planning' strategies should be based on a close analysis of "the difficulties of […] [the] processes as well as to the range of contexts and conditions within which participation takes place" (Watson, 2014, p. 63), which would save them (the 'negotiated planning' strategies) from the traps of an idealized image of collaborative planning based on the Habermasian model (Tewdwr-Jones, Allmendinger, 1998).

The reflections developed here regarding the role of urban commons in shaping urban planning strategies aimed to render explicit the relevance and importance of establishing methods intending to examine the actual practices of citizens while making decisions. Within the framework if this endeavor to better grasp the role of urban commons in shaping urban planning strategies a concept that is of great importance is that of negotiated planning, which "focuses less on normative expressions of how planning should be (i.e., informed by evidence and

participation) and more on the actual practices evident in cities" (Cirolia, Berrisford, 2017). The shift from 'collaborative' toward 'negotiated planning' is related to the intensification of the interest in commoning practices, and in the actual "actors and power dynamics […] involved," as well as in "the 'virtuous cycle' of planning, infrastructure, and land" (Cirolia, Berrisford, 2017, p. 77). At the core of current debates are the role of urban commons in spatial planning decision-making, and the exploration of how we could combine the use of advanced digital tools such as digital twins and civic-oriented participatory design methods. Despite the aspirations of urban scale digital twins to enhance the participation of citizens in the decision-making processes relayed to urban planning strategies, the fact that they are based on a limited set of variables and processes makes them problematic. It is of pivotal importance to shape methodological tools offering the possibility to develop new forms of social advocacy around Big Data, bringing together the reflection on smart cities and the debates around urban commons.

## ENDNOTES

1. The research project was supported by the Hellenic Foundation for Research and Innovation (H.F.R.I.) under the "3rd Call for H.F.R.I. Research Projects to support Post-Doctoral Researchers" (Project Number: 7833)
2. Engineering and Technology (IET) (2022). *The Apollo Protocol: unifying digital twins across sectors*. London: Institution of Engineering and Technology (IET). Available at: https://www.theiet.org/media/10434/the-apollo-protocol-unifying-digital-twins-across-sectors.pdf (Accessed on 1 April 2022)
3. Engineering and Technology (IET) (2019). *Digital Twins for the Built Environment: An Introduction to the Opportunities, Benefits, Challenges and Risks.* London: Institution of Engineering and Technology (IET). Available at: https://www.theiet.org/media/8762/digital-twins-for-the-built-environment.pdf (Accessed on 1 April 2022)
4. Department for Business and Trade (DBT) (2023). 'AccuCities: 3D Model of London in SimScale wind comfort simulation'. Available at: https://www.great.gov.uk/international/trade/case-study/3474/3d-model-of-london-in-simscale-wind-comfort-simula/(Accessed on 1 April 2022)

## REFERENCES

Batty, M. (2018). 'Digital Twins', *Environment and Planning B: Urban Analytics and City Science*, 45(5), doi: https://doi.org/10.1177/2399808318796416

Blair, G. S. (2021). 'Digital twins of the natural environment', *Patterns*, 2(10), doi: https://doi.org/10.1016/j.patter.2021.100359

Bolton, A., Butler, L., Dabson, I., Enzer, M., Evans, M., Fenemore, T., Harradence, F., et al. (2018). *Gemini Principles. (CDBB_REP_006),* doi: https://doi.org/10.17863/CAM.32260

Caprari, G., Castelli, G., Montuori, M., Camardelli, M., Malvezzi, R. (2022). 'Digital Twin for Urban Planning in the Green Deal Era: A State of the Art and Future Perspectives', *Sustainability*, 14(1): 6263, doi: https://doi.org/10.3390/su14106263

Castells, M. (1989). *The Informational City: Information Technology, Economic Restructuring and the Urban-Regional Process*. Oxford: Basil Blackwell.

Castells, M. (2009). *Communication Power*. Oxford: Oxford University Press.

Castells, M. (2010). Communication Power; Castells, "Globalisation, Networking, Urbanisation: Reflections on the Spatial Dynamics of the Information Age', *Urban Studies*, 47(13): 2737-2745.

Castells, M. (2010). *The Rise of the Network Society. Second edition with a new preface*. Chichester, West Sussex: Wiley-Blackwell.

Castells, M. (2021). 'From Cities to Networks: Power Rules', *Journal of Classical Sociology*, 21(3-4): 260–262, doi: https://doi.org/10.1177/1468795X21102205

Charitonidou, M. (2022) ,Urban scale digital twins in data-driven society: Challenging digital universalism in urban planning decision-making', *International Journal of Architectural Computing*, 20(2): 238-253, doi: https://doi.org/10.1177/14780771211070005

Cirolia, L. R., Berrisford, S. (2017). ,Negotiated planning: Diverse trajectories of implementation in Nairobi, Addis Ababa, and Harare', *Habitat International*, 59: 71–79, doi: https://doi.org/10.1016/j.habitatint.2016.11.005

Deren, L., Wenbo Y., Zhenfeng S. (2021). 'Smart city based on digital twins', *Computational Urban Science*, 1(4), 1, doi: https://doi.org/10.1007/s43762-021-00005-y

Ferré-Bigorra, J., Casals, M., Gangolells, M. (2022). 'The adoption of urban digital twins', *Cities*, 131, 103905, doi: https://doi.org/10.1016/j.cities.2022.103905

Hatcher, J. (2022). 'Digital twins can help sustainability', *Smart Building Magazine*. URL: https://smartbuildingsmagazine.com/news/digital-twins-can-help-sustainability

Hernández-Morales, A. (2022). 'Barcelona bets on 'digital twin' as future of city planning', *Politico*. Available at: https://www.politico.eu/article/barcelona-digital-twin-future-city-planning/ (accessed 1 June 2022)

Joss, S., Sengers, F., Schraven, D., Caprotti, F., Dayot, Y. (2019). 'The Smart City as Global Discourse: Storylines and Critical Junctures across 27 Cities', *Journal of Urban Technology*, 26(1), doi: https://doi.org/10.1080/10630732.2018.1558387

Kaufmann, V. (2003). *Re-Thinking Mobility: Contemporary Sociology*. London; New York: Routledge.

Loukissas, Y. A. (2019). *All Data Are Local: Thinking Critically in a Data-Driven Society*. Cambridge, Mass.: The MIT Press.

Nochta, T., Wan, L., Schooling, J. M., Parlikad, A. K. (2021). 'A Socio-Technical Perspective on Urban Analytics: The Case of City- Scale Digital Twins', *Journal of Urban Technology*, 28(1-2): 263-287, doi: https://doi.org/10.1080/10630732.2020.1798177

Raes, L., Michiels, P., Adolphi, T., Tampere, C., Dalianis, A., McAleer, S., Kogut, P. (2022). 'DUET: A Framework for Building Secure and Trusted Digital Twins of Smart Cities', *IEEE Internet Computing*, 26(3): 43–50, doi: https://doi.org/10.1109/MIC.2021.3060962

Sheller, M. (2018). *Mobility Justice: The Politics of Movement in an Age of Extremes*. London: Verso.

Sheller, M. (2020). 'Mobility justice in urban studies', in Jensen, O. B., Lassen, C., Kaufmann, V., Freudendal-Pedersen, M., Gøtzsche Lange, I. S. eds., *Handbook of Urban Mobilities*. London; New York: Routledge, 13-22.

Tewdwr-Jones, M., Allmendinger, P. (1998). 'Deconstructing communicative rationality: A critique of Habermasian collaborative Planning', *Environment and Planning A: Economy and Space*, 30(11): 1975–1989, doi: https://doi.org/10.1068/a301975

Tomko, M., Winter, S. (2019). 'Beyond digital twins – A commentary', *Environment and Planning B: Urban Analytics and City Science*, 46(2), doi: https://doi.org/10.1177/2399808318816992

Watson, V. (2014). 'Co-production and collaboration in planning: The difference', *Planning Theory & Practice,* 15(1), doi: https://doi.org/10.1080/14649357.2013.866266

White, A. (2016). Manuel Castells's trilogy the information age: economy, society, and culture, Information, *Communication & Society*, 19(12), doi: https://doi.org/10.1080/1369118X.2016.1151066

White, G., Zink, A., Codecá, L., Clarke, S. (2021). 'A digital twin smart city for citizen feedback', *Cities*, 110, doi: https://doi.org/10.1016/j.cities.2020.103064

Wildfire, C. (2018). 'How can we spearhead city-scale digital twins?', *Infrastructure Intelligence*, 9. Available at: http://www.infrastructure-intelligence.com/article/may-2018/how-can-we-spearhead-city-scale-digital-twins (accessed 3 April 2021).